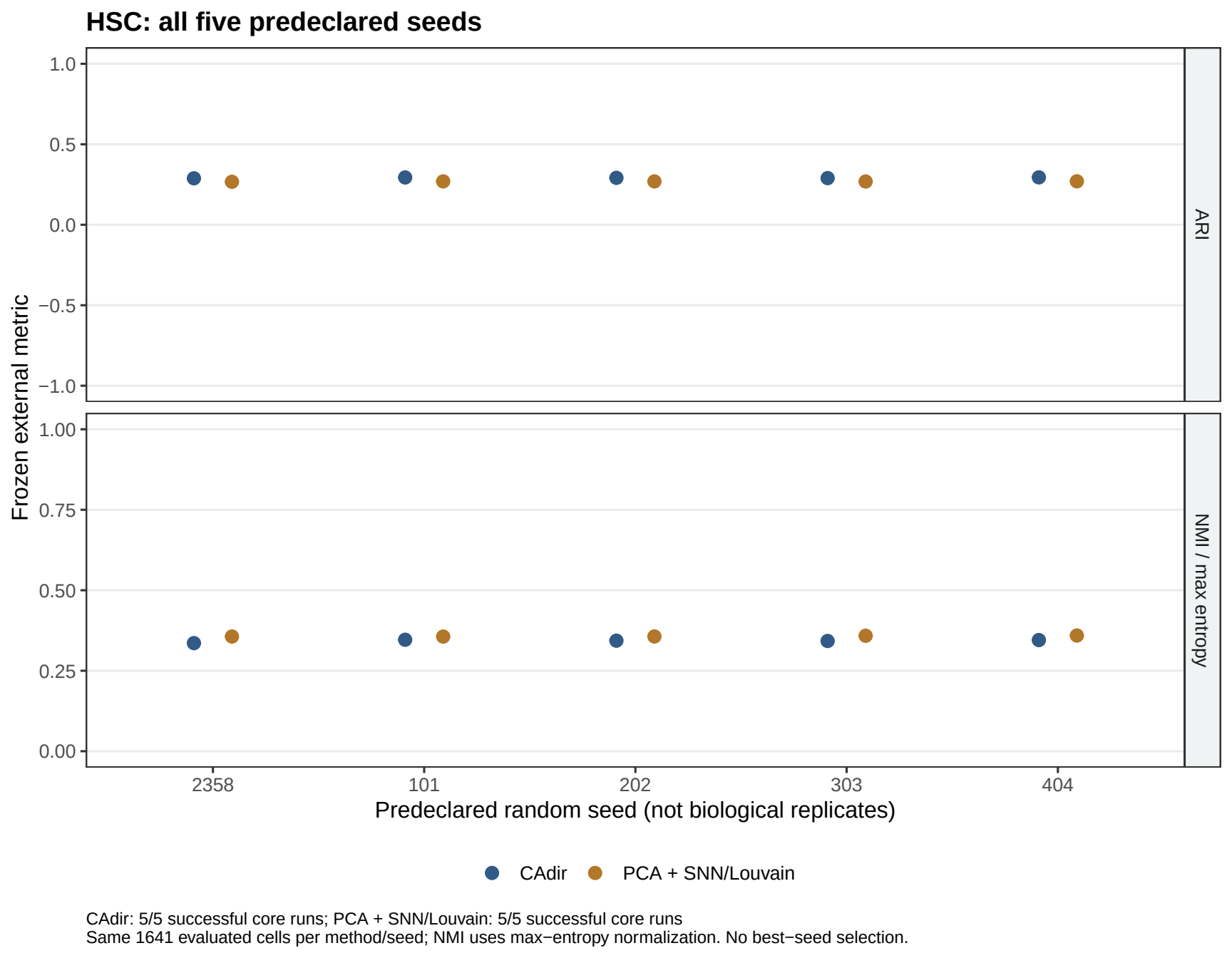

CAdir: 5/5 successful core runs; PCA + SNN/Louvain: 5/5 successful core runs
Same 1641 evaluated cells per method/seed; NMI uses max−entropy normalization. No best−seed selection.